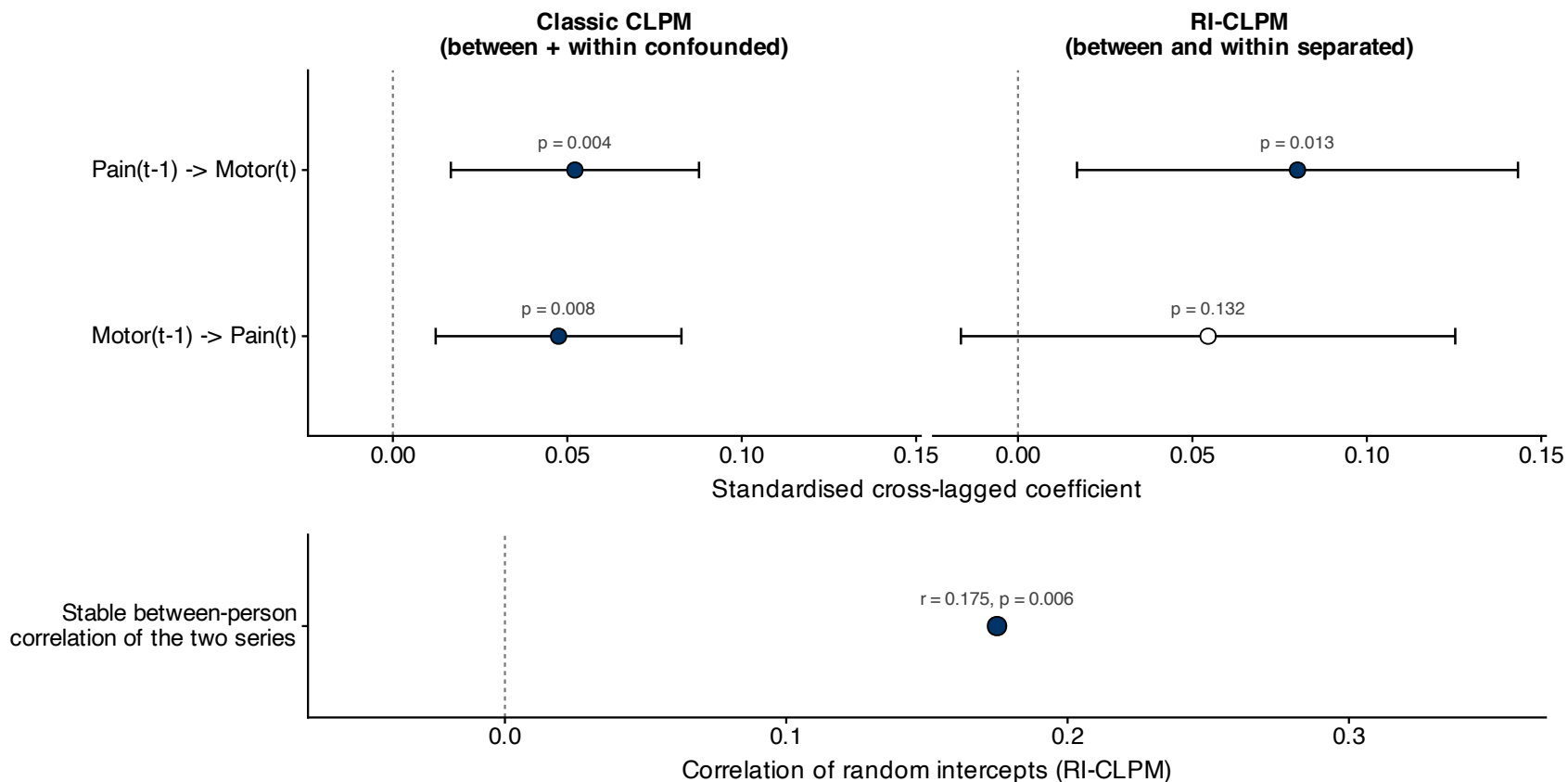


**Figure 2. Direction of the pain and motor severity relationship**



Left panels: the classic cross-lagged panel model makes both directions significant. Right: once stable between-person differences are separated by random intercepts, the motor-to-pain path is null and only a modest pain-to-motor path remains, which does not survive the robustness checks reported in the text. Bottom: the two series covary stably across people. Filled circles denote  $p < 0.05$ . Model fit favours the RI-CLPM (CFI 0.960 vs 0.877; RMSEA 0.067 vs 0.114).